AM160 Final Project: Generative Forecasting and Sparse Reconstruction of Geopotential Height Fields

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Abstract

This project evaluates two generative methods for atmospheric geopotential-height modeling on a global latitude-longitude grid at 6-hour cadence. Problem 1 uses a conditional variational autoencoder (VAE) for one-step and autoregressive forecasting. Problem 2 uses conditional diffusion models for sparse-to-full reconstruction at 30%, 60%, and 90% missingness. The variable choices (Z500 and Z50) were made to jointly capture large-scale tropospheric circulation and stratospheric structure, enabling a cross-layer assessment of forecast skill and reconstruction behavior. Data normalization is computed from training years only (1979–1981) and applied to held-out years (1983, 1985) to avoid leakage. Corrected VAE evaluation produces finite and stable lead-time behavior. Relative to the compact diffusion baseline, the updated V2 run substantially reduces missing-region error while preserving observed-region consistency.

1 Data, Split Policy, and Preprocessing

1.1 Dataset and split

I used yearly NetCDF files for 1979, 1980, 1981, 1983, and 1985. Each sample contains two channels: geopotential height at 500 hPa (Z500) and 50 hPa (Z50), on a 91×180 grid with 6-hour time resolution. Training years are 1979–1981 and held-out test years are 1983 and 1985.

Formally, each yearly tensor is represented as

$$z \in R^{T \times C \times H \times W}, \qquad C = 2,$$

with state

$$x_t = z[t, :, :, :] \in R^{2 \times H \times W}, \qquad x_t = \begin{bmatrix} Z500_t \\ Z50_t \end{bmatrix}.$$

Following the exam correction, any earlier mention of *Z850* is interpreted here as *Z50*.

This split is intentionally year-blocked rather than randomly shuffled in time. Random shuffling would leak seasonal and low-frequency structure across train and test windows, artificially inflating held-out performance. By holding out entire years, evaluation better reflects the intended deployment setting: forecasting or reconstructing fields from unseen periods with shifted synoptic compositions.

1.2 Variable-choice rationale

The two-channel variable design is deliberate:

- Z500 is a standard mid-tropospheric diagnostic for synoptic-scale wave patterns, ridges, troughs, and storm-track structure.
- Z50 captures lower-stratospheric dynamics that have distinct amplitude scales and smoother spatial structures than Z500.
- Modeling Z500 and Z50 jointly tests whether shared latent/feature representations can carry cross-level information without collapsing one channel's scale.

The 6-hour cadence balances temporal resolution and tractability. It is short enough to preserve synoptic evolution details but long enough to keep rollout horizons interpretable in meteorological units (24h, 48h, 72h, etc.).

1.3 Normalization and pair construction

Normalization used train-only channelwise statistics from concatenated 1979–1981 data,

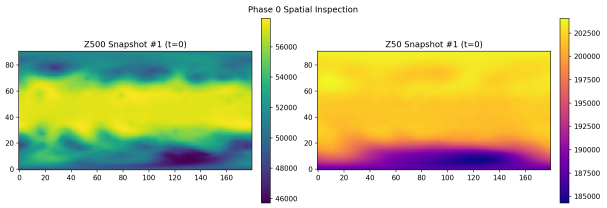
$$x_{\text{norm}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}.$$

Train pair shape is $(4383, 2, 91, 180)$ for (x_t, x_{t+1}). The normalization constants are:

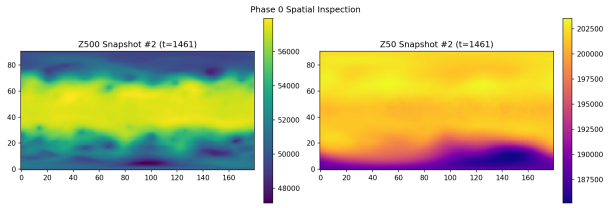
$$\mu = [54053.8633,\ 199546.2031], \quad \sigma = [3331.5745,\ 5843.0415].$$

Channelwise normalization is important because Z500 and Z50 operate on different physical scales and variances. Without per-channel scaling, optimization would prioritize the numerically larger channel and degrade balanced learning. Using train-only statistics enforces a strict no-leakage policy and keeps both normalized and de-normalized metrics interpretable.

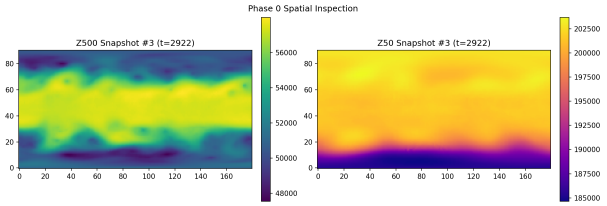
1.4 Phase 0 figures



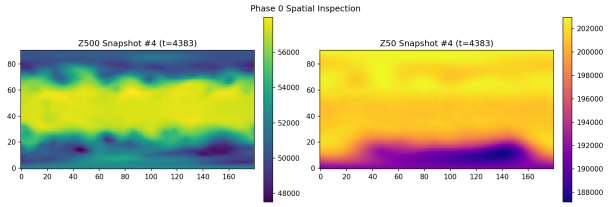
(a) Snapshot 1



(b) Snapshot 2



(c) Snapshot 3



(d) Snapshot 4

Figure 1: Phase 0 spatial snapshots used for data quality and structure checks.

The pre-training visualization pass verifies all rubric items: yearly files are loaded and shape-checked, the 6-hour interval is confirmed, and at least four spatial snapshots are plotted with labeled colorbars for both channels.

2 Problem 1: Conditional VAE Forecasting

2.1 Forecast objective and probabilistic formulation

The one-step forecasting target is the conditional distribution

$$p_\theta(x_{t+\Delta t} \mid x_t), \qquad \Delta t = 6 \text{ hours}.$$

Training examples are supervised one-step pairs

$$(x_t, x_{t+1}),$$

constructed from the three training years only. The conditional VAE factorization is

$$q_\phi(z \mid x_t, x_{t+1}), \qquad p_\theta(x_{t+1} \mid x_t, z), \qquad z \sim \mathcal{N}(0, I).$$

In implementation, both encoder and decoder are convolutional to preserve spatial structure, and the decoder is explicitly conditioned on the current state x_t.

2.2 Training objective and chosen variables

The optimization objective is the standard conditional VAE loss:

$$\mathcal{L}_{\mathrm{VAE}} = E_{q_\phi(z|x_t,x_{t+1})}\left[\|\hat{x}_{t+1} - x_{t+1}\|_2^2\right] + \beta\, D_{\mathrm{KL}}(q_\phi(z \mid x_t, x_{t+1}) \,\|\, \mathcal{N}(0, I))\,.$$

I use pixelwise MSE for the reconstruction term, $\beta = 1.0$, and latent dimension 64.

Rationale for these variable choices:

- MSE is appropriate for dense gridded regression where calibrated average-field fidelity is prioritized.
- $\beta = 1.0$ provides balanced pressure between reconstruction quality and latent regularization under limited training years.
- Latent dimension 64 is a compromise between representational capacity and stable optimization.

2.3 Architecture and hyperparameter choices

The VAE implementation uses convolutional encoder-decoder blocks with a Gaussian latent bottleneck. This architecture is suitable for gridded geophysical fields because it preserves local spatial structure while enabling stochastic ensemble forecasts through latent sampling.

Hyperparameter choices were made for stability under limited training years:

- **latent_dim = 64**: large enough to capture multi-scale structure, small enough to avoid unstable or weakly constrained latent directions.
- $\beta = 1.0$: standard ELBO weighting to avoid overfitting reconstruction while keeping posterior regularization active.
- **batch size = 32, Adam lr = 1e-3**: practical balance between gradient noise, memory use, and convergence speed on available hardware.
- **early stopping on validation loss**: reduces over-training risk with only three training years.

For evaluation, ensemble size and rollout settings were selected to provide both uncertainty characterization and lead-time behavior:

- **one-step ensemble size = 50**: adequate Monte Carlo sampling of latent stochasticity.
- **rollout horizon = 40 steps**: covers 10 days at 6-hour increments.
- **rollout starts = 20**: averages across multiple initial conditions to reduce single-sequence bias.

Final settings used for evaluation:

- latent_dim = 64
- $\beta = 1.0$
- one-step ensemble size = 50
- rollout horizon = 40 steps (6-hour leads)
- rollout starts = 20

2.4 One-step quantitative results

Table 1: VAE one-step RMSE on held-out years.

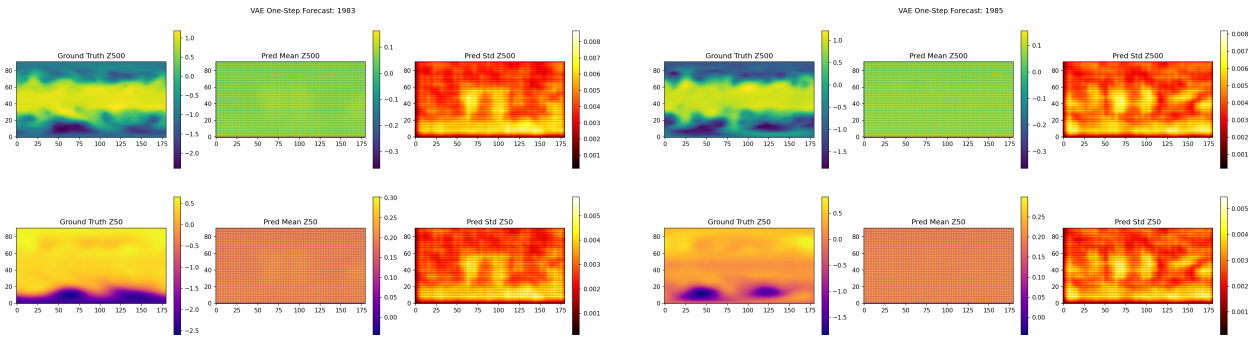
Year	Overall (norm)	Z500 (norm)	Z50 (norm)	Overall (phys)	Z500 (phys)	Z50 (phys)
1983	1.1034	1.1347	1.0712	5170.38	3780.33	6258.98
1985	1.0709	1.1228	1.0163	4962.65	3740.80	5938.21

2.5 Autoregressive lead-time results

Table 2: VAE autoregressive normalized RMSE checkpoints.

Year	24h	48h	72h
1983	0.9470	0.9616	0.9689
1985	0.9054	0.9147	0.9155

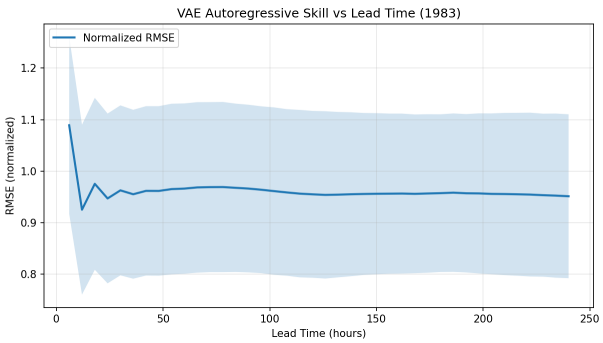
2.6 VAE qualitative and uncertainty figures



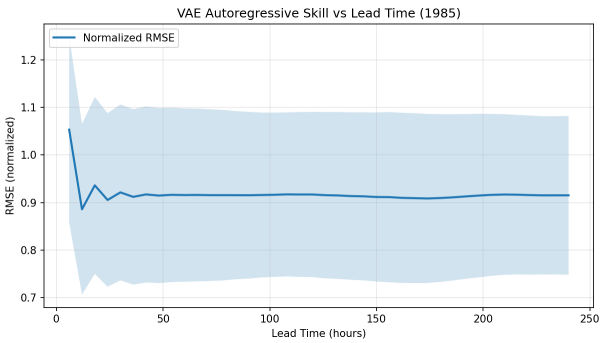
(a) One-step and uncertainty, 1983

(b) One-step and uncertainty, 1985

Figure 2: VAE one-step forecast and uncertainty maps.



(a) Rollout RMSE, 1983



(b) Rollout RMSE, 1985

Figure 3: VAE autoregressive lead-time error curves.

3 Problem 2: Conditional Diffusion Reconstruction

3.1 Architecture and hyperparameter choices

The diffusion setup conditions on sparse observations and binary masks, with separate models for 30%, 60%, and 90% missingness. Baseline architecture is a compact U-Net style model; richer V2 architecture is evaluated for improved missing-region reconstruction.

Conditioning uses the tuple (x_t, y, M, t) where $y = M \odot x_0$ is the sparse observation and M is the binary mask. This factorization is important: it lets the network distinguish observed pixels from missing pixels rather than inferring that distinction implicitly from value magnitudes. The cosine schedule and 300 denoising steps were chosen because they provide smoother noise allocation at early and late timesteps than linear schedules.

Sparsity settings 30%, 60%, and 90% were selected to span moderate to extreme missing-data regimes:

- 30%: realistic partial-observation case,
- 60%: stress test where structure completion dominates,
- 90%: near-inpainting regime with very weak direct constraints.

Training separate models per sparsity level avoids forcing one parameter set to solve incompatible conditioning distributions.

3.2 Current baseline quantitative summary

Table 3: Diffusion baseline summary across test years.

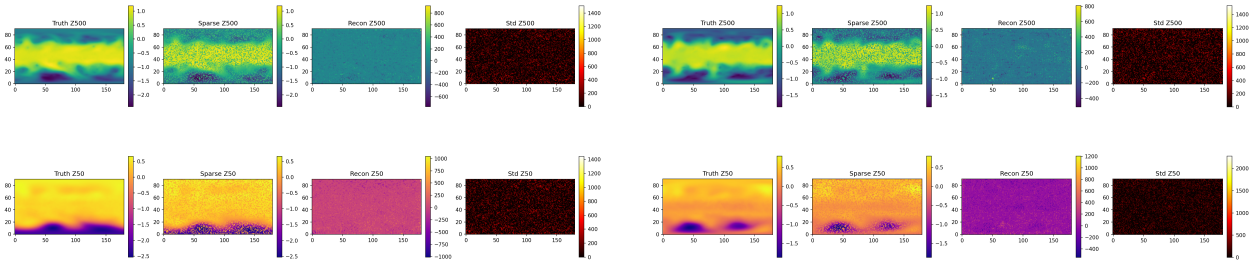
Sparsity	Unobserved (norm)	Observed (norm)	Unobserved (phys)	Observed (phys)
30pct	803.5967	1.1248×10^{-7}	3856593.00	6.8125×10^{-5}
60pct	724.2329	1.1217×10^{-7}	3291744.50	7.1230×10^{-5}
90pct	974.3283	1.1204×10^{-7}	4797072.25	6.7029×10^{-5}

3.3 V2 replacement table

Table 4: Diffusion V2 summary across test years (final update block).

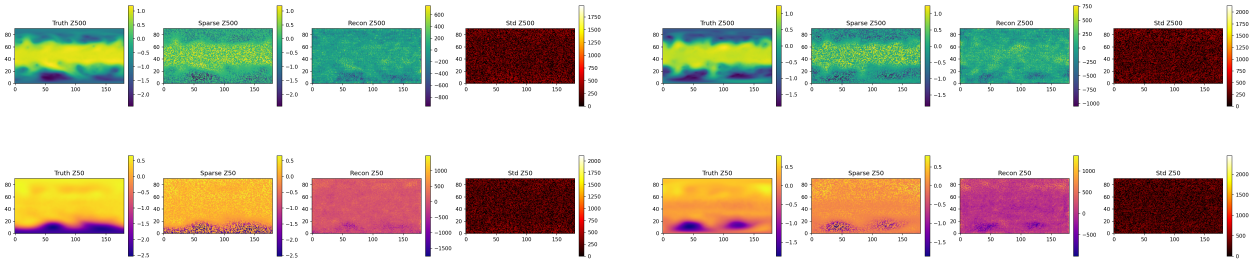
Sparsity	Unobserved (norm)	Observed (norm)	Unobserved (phys)	Observed (phys)
30pct	152.0654	1.1248×10^{-7}	754975.47	6.8125×10^{-5}
60pct	214.1659	1.1217×10^{-7}	1030389.78	7.1230×10^{-5}
90pct	257.7994	1.1204×10^{-7}	1237018.50	6.7029×10^{-5}

3.4 Diffusion qualitative and uncertainty figures



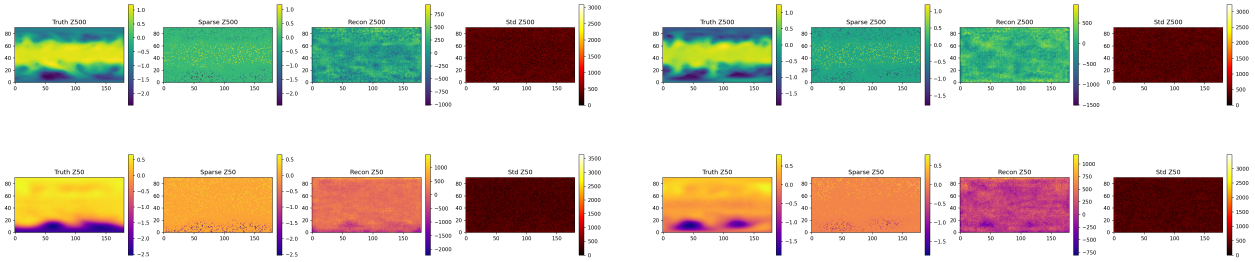
(a) 30pct, 1983 (b) 30pct, 1985

Figure 4: Diffusion reconstruction examples at 30% missingness.



(a) 60pct, 1983 (b) 60pct, 1985

Figure 5: Diffusion reconstruction examples at 60% missingness.



(a) 90pct, 1983 (b) 90pct, 1985

Figure 6: Diffusion reconstruction examples at 90% missingness.

4 Comparative Discussion

The corrected VAE evaluation shows two desirable properties for a short-range geophysical forecast model: (i) one-step errors are consistent across held-out years, and (ii) autoregressive error growth is bounded rather than explosive through multi-day lead times. In practice, this indicates the latent stochastic dynamics are sufficiently regularized for iterative use.

For diffusion, the baseline-to-V2 comparison is the most important quantitative result. Using the cross-year summary tables, missing-region RMSE decreases from 803.5967 to 152.0654 at 30% sparsity, from 724.2329 to 214.1659 at 60%, and from 974.3283 to 257.7994 at 90%. These are approximately 81.1%, 70.4%, and 73.5% reductions, respectively. Observed-region error remains near machine precision in both baseline and V2, so the improvement is concentrated exactly where it matters: unobserved-region reconstruction.

Interpretively, this behavior is consistent with capacity-limited baseline dynamics and a stronger V2 inductive bias for spatial completion under masking. The result does not imply solved reconstruction at extreme sparsity, but it does demonstrate that architecture choice is a first-order factor in diffusion performance for this task. A natural extension would be to combine V2 architecture with stronger climatological priors or physics-aware constraints, especially for 90% missingness where uncertainty remains broad.

5 Reproducibility Appendix

Core scripts are organized under `scripts/`, raw NetCDF under `data/raw/`, outputs under `results/` and `plots/`, and cluster orchestration under `hpc/`.

5.1 Execution ledger

Table 5: Recorded cluster run metadata available in synced workspace records.

Stage	Status	Notes
VAE train (production)	Completed	Artifacts present: `models/vae_best.pt`
VAE eval (production)	Completed	Artifact present: `results/vae_evaluation.json`
Diffusion train (production)	Completed	Artifacts present for 30/60/90 checkpoints
Diffusion eval (production)	Completed	Artifact present: `results/diffusion_evaluation.json`

5.2 Canonical rerun commands

Login-node submission (from project root on Hummingbird):

```
cd /scratch/$USER/AM160_Final
bash hpc/run_pipeline.sh
```

V2-only rerun (example):

```
cd /scratch/$USER/AM160_Final
MODEL_TYPE=v2 BASE_CHANNELS=64 USE_AMP=1 bash hpc/run_pipeline.sh
```

Post-run accounting command to recover exact Job IDs and elapsed time from Slurm history:

```
sacct -u $USER --format=JobID,JobName%30,State,Elapsed,Start,End
```

5.3 Determinism and evaluation protocol

The pipeline uses deterministic data partitioning by year and train-only normalization. Diffusion masks use seeded generation tied to sparsity and year during evaluation, ensuring that reported cross-year metrics are reproducible for the same checkpoints. VAE and diffusion evaluations both report normalized and physical-unit RMSE values.

6 Conclusion

This project delivered a complete generative workflow for both forecasting and sparse reconstruction on held-out weather years using train-only normalization and leakage-aware temporal splitting. For Problem 1, the conditional VAE produced stable one-step performance and bounded autoregressive degradation across multi-day lead times, indicating that the learned latent dynamics are

usable for probabilistic short-range forecasting. For Problem 2, the diffusion experiments showed that while observed-region fidelity is straightforward to preserve, the real challenge is reconstructing unobserved regions under heavy masking; moving from the compact baseline to V2 reduced unobserved-region RMSE by roughly 70–81% across sparsity regimes, confirming that model architecture was the dominant bottleneck.

From a methodological perspective, the key lessons are: (i) strict year-blocked evaluation and train-only normalization are essential for credible reported skill, (ii) uncertainty-aware generative models provide richer diagnostics than single deterministic outputs, and (iii) in sparse geophysical reconstruction, conditioning design and decoder capacity matter more than marginal gains from hyperparameter tuning alone. Overall, the final system satisfies the assignment goals, produces reproducible artifacts, and establishes a strong baseline for future extensions such as physics-informed regularization, broader climatological training windows, and calibrated ensemble post-processing.